VHDL code for Comparator

This VHDL project presents a simple VHDL code for a comparator which is designed and implemented in Verilog before. Full VHDL code together with test bench for the comparator is provided.
The design for the comparator based on the truth table and K-map are already presented here. There are two 2-bit inputs A and B to be compared. Three output signals are A_less_B (1 if A < B, else 0), A_equal_B (1 if A=B, else 0), and A_greater_B (1 if A>B, else 0). The minimized expressions obtained from K-Map tables for the outputs are used for VHDL coding of the comparator.

VHDL code for the comparator:

```
-- fpga4student.com: FPGA projects, Verilog projects, VHDL projects
-- VHDL project: VHDL code for a comparator
-- A comparator with 2 2-bit input A and B, and three outputs
-- A less than B, A equal B, and A greater than B
-- The comparator is designed by using truth table, K-Map.
-- The final output expressions are used for VHDL coding
library IEEE;
use IEEE.std_logic_1164.all;
entity comparator_VHDL is
port (
 A,B: in std_logic_vector(1 downto 0); -- two inputs for comparison
 A_less_B: out std_logic; -- '1' if A < B else '0'
 A_equal_B: out std_logic;-- '1' if A = B else '0'
 A_greater_B: out std_logic-- '1' if A > B else '0'
 );
end comparator_VHDL;
architecture comparator_structural of comparator_VHDL is
signal tmp1,tmp2,tmp3,tmp4,tmp5, tmp6, tmp7, tmp8: std_logic;
-- temporary signals
begin
 -- A_equal_B combinational logic circuit
 tmp1 <= A(1) xnor B(1);
 tmp2 <= A(0) xnor B(0);
 A_equal_B <= tmp1 and tmp2;
 -- A_less_B combinational logic circuit
 tmp3 <= (not A(0)) and (not A(1)) and B(0);
 tmp4 <= (not A(1)) and B(1);
 tmp5 <= (not A(0)) and B(1) and B(0);
 A_less_B <= tmp3 or tmp4 or tmp5;
 -- A_greater_B combinational logic circuit
 tmp6 <= (not B(0)) and (not B(1)) and A(0);
 tmp7 <= (not B(1)) and A(1);
 tmp8 <= (not B(0)) and A(1) and A(0);
 A_greater_B <= tmp6 or tmp7 or tmp8;
end comparator_structural;
```

VHDL testbench code for the comparator:

```
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-- VHDL project: VHDL code for a comparator
-- Testbench VHDL code for comparator
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;
use ieee.numeric_std.all;
ENTITY tb_comparator_VHDL IS
END tb_comparator_VHDL;

ARCHITECTURE behavior OF tb_comparator_VHDL IS

    -- Component Declaration for the comparator in VHDL

    COMPONENT comparator_VHDL
    PORT(
         A : IN  std_logic_vector(1 downto 0);
         B : IN  std_logic_vector(1 downto 0);
         A_less_B : OUT  std_logic;
         A_equal_B : OUT  std_logic;
         A_greater_B : OUT  std_logic
        );
    END COMPONENT;
   --Inputs
   signal A : std_logic_vector(1 downto 0) := (others => '0');
   signal B : std_logic_vector(1 downto 0) := (others => '0');
  --Outputs
   signal A_less_B : std_logic;
   signal A_equal_B : std_logic;
   signal A_greater_B : std_logic;
BEGIN

 -- Instantiate the comparator in VHDL
   uut: comparator_VHDL PORT MAP (
          A => A,
          B => B,
          A_less_B => A_less_B,
          A_equal_B => A_equal_B,
          A_greater_B => A_greater_B
        );

   -- Stimulus process
   stim_proc: process
   begin
  -- create test cases for A_less_B
  for i in 0 to 3 loop
          A <= std_logic_vector(to_unsigned(i,2));
          B <= std_logic_vector(to_unsigned(i+1,2));
          wait for 20 ns;
      end loop;
  -- create test cases for A_greater_B
   for i in 0 to 3 loop
          A <= std_logic_vector(to_unsigned(i+1,2));
          B <= std_logic_vector(to_unsigned(i,2));
          wait for 20 ns;
      end loop;
  -- create test cases for A_equal_B
  for i in 0 to 3 loop
          A <= std_logic_vector(to_unsigned(i,2));
          B <= std_logic_vector(to_unsigned(i,2));
          wait for 20 ns;
      end loop;
      wait;
   end process;
```
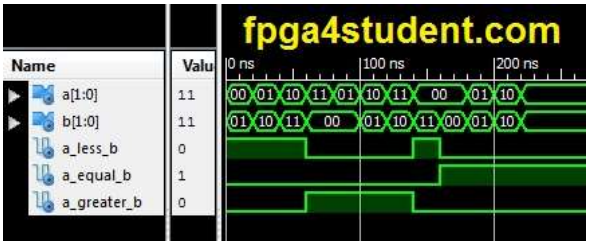

```
END;
```

Simulation waveform for the comparator in VHDL:



The simulation waveform above shows the correct operations of the comparator. Another 8-bit comparator design in VHDL is presented here.

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